

Complex Variables, Laplace and Z-transformations (Mid-Summer '23-24)

Syllabus for Midterm exam – Laplace transformation; Inverse Laplace transformation; Application of Laplace Transformation; System of ODE using Laplace transformation; Complex number; Complex function and its differentiation (Cauchy-Riemann equations).

Advanced Engineering Mathematics – Erwin Kreyszig - 10th edition.

Topics	Sub topics and Example	Exercise
Laplace Transformation (Chapter- 6)	Definition (P-203) Derivation of formula: Example # 1- 4 Table-6.1 (P-207)	P-210 Ex # 1 -6 P-210 Ex # 9-16 (Using definition) P-251 Ex # 11-13,17,19
	Linearity of Laplace transform (P-206), First Shifting Theorem (P- 208) P-208 Example # 5	P- 211 Ex # 33-36
Inverse Laplace Transformation (Chapter- 6)	Definition (P-205) Theorem 2 (P-208) Example #5	P- 211 Ex # 25-32 (Using formula and partial fraction) P- 211 Ex # 37-41, 43,44 P-251 Ex # 20,22-25,27,28
Application of Laplace Transformation (Chapter- 6)	Derivative formula of Laplace transformation (P-211) P-214 Example # 4,5	P- 216 Ex # 1-11,16,17 P-251 Ex # 29,30
	Some Physical Problem	Notes will be provided
Unit Step Function (Chapter 6)	Unit Step Function (P-217) Time Shifting (3,4,4*) (P-219) P-220 Example # 1,2 Dirac's Delta Function (P-225)	P-223 Ex # 2-17 P-230 Ex # 3-8,11,12 P-251 Ex # 14-16,21,26 P-252 Ex # 31-33
System of ODE using Laplace Transformation (Chapter- 6)	P-242 Example # 1,3	P- 246 Ex # 3,6,8,9,11,12 P-252 Ex # 34-36

Complex Analysis with Applications – Dennis G. Zill.

Topics	Subtopics and Example	Exercise
Complex Numbers (Chapter- 1)	Complex numbers and their properties (P-2) Conjugate (P-4) (P-4) Example # 1,2,3	P-7 Ex # 1,2,3-16 P-8 Ex # 21, 25, 27-34, 48
Complex Plane (Chapter- 1)	Complex Plane (P-10) Modulus (P-10), Properties (P-11), Distance (P-11), Inequalities (P-12) (P-11) Example # 2	P- 14 Ex # 6(a), 9-12, 27-29
Polar Form of Complex Numbers (Chapter- 1)	Polar Form (P-16), Principal Arguments (P-17), Arguments, Multiplication and Division (P-18), Integer Power of z, De Moivre's Formula (P- 20) P-17 Example # 1,2,3,4	P- 21 Ex # 1-8, 23
Powers and Roots (Chapter-1)	Roots (P-24), Principal n-th Roots (P-25) (P-24) Example # 1,2	P-8 Ex# 35,36 P-26 Ex # 1,2,7-10,17
Differentiability and Analyticity (Chapter- 3)	Derivatives (P-142) Analytic Functions (P-145) Cauchy-Reimann Equations (C-R) (P-152) Theorem-3.4, Formula of $f'(z)$ (P-152) (P-153) Example # 1,2 Theorem-3.5 (P-154) (P-155) Example # 3 Polar Coordinates (P-156)	Verify C-R Equations and find $f'(z)$: P-157 Ex#1-12,15,16,25, $f(z) =$ $z^5, \frac{1}{z^7}, 2e^z, e^{3z}, e^{-z}, 2z^3 + 6z$
Mappings (Chapter- 2)	Linear Mappings (2.3) (Page-69,73,74) Example # 1,4,5	(Page-76,77) Ex # 7-16
Integration in the Complex Plane (Chapter- 5)	Complex Integrals (5.2) (Page-250-252) Example # 1-3	(P-254-256) Ex # 1-12,14-24

Exercise 1.1

I. Find the Laplace transformation (1-6) using definition:

- | | |
|-------------------|-----------------------|
| 1. $L\{e^{at}\}$ | 4. $L\{t^2 e^{at}\}$ |
| 2. $L\{\sin 2t\}$ | 5. $L\{e^{at} f(t)\}$ |
| 3. $L\{\cos 3t\}$ | 6. $L\{\cos at\}$ |

II. Find the Laplace transformation of the following functions (7-26), assuming a, b, ω, π as constants:

- | | |
|--|----------------------------------|
| 7. $f(t) = 3t + 12$ | 17. $f(t) = e^{2t} \sinh t$ |
| 8. $f(t) = (a - bt)^2$ | 18. $f(t) = e^{-t} \sinh 4t$ |
| 9. $f(t) = \cos \pi t$ | 19. $f(t) = \sinh t \cos t$ |
| 10. $f(t) = \cos^2 \omega t$ | 20. $f(t) = 12te^{-3t}$ |
| 11. $f(t) = \sin^2\left(\frac{1}{2}\pi t\right)$ | 21. $f(t) = t^2 e^{-3t}$ |
| 12. $f(t) = 5 \cosh 2t - 3 \sinh t$ | 22. $f(t) = t \cos t + \sin t$ |
| 13. $f(t) = e^{-t} (\cos 4t - 2 \sin 4t)$ | 23. $f(t) = e^{2t} \sinh t$ |
| 14. $f(t) = ke^{-at} \cos \omega t$ | 24. $f(t) = e^{-t} \sinh 4t$ |
| 15. $f(t) = 0.5e^{-4.5t} \sin 2\pi t$ | 25. $f(t) = e^t (t^2 + \sin 3t)$ |
| 16. $f(t) = t^5 e^{2t}$ | 26. $f(t) = e^{-3t} (t + 4)^2$ |

Additional problem to solve

- | | |
|---|---|
| 1. $f(t) = (t^2 + 1)^2$ | Ans: $\frac{24}{s^5} + \frac{4}{s^3} + \frac{1}{s}$ |
| 2. $f(t) = e^{-3t} + 5 \cosh t$ | Ans: $\frac{1}{s+3} + \frac{5s}{s^2-1}$ |
| 3. $f(t) = (\sin t - \cos t)^2 + \cos^2 3t$ | Ans: $\frac{3}{2s} - \frac{2}{s^2+4} + \frac{1}{2} \left(\frac{s}{s^2+36} \right)$ |
| 4. $f(t) = 3 - e^{-t} \cos 2t + t \sinh 3t$ | Ans: $\frac{3}{s} - \frac{s+1}{s^2+2s+5} + \frac{6s}{(s^2-9)^2}$ |
| 5. $f(t) = t \cos t - e^{-3t} \sin 2t + t^7 e^{5t}$ | Ans: $-\frac{1-s^2}{(s^2+1)^2} - \frac{2}{(s+3)^2+4} + \frac{5040}{(s-5)^8}$ |
| 6. $f(t) = e^{2t} t + e^t \cosh 3t$ | Ans: $\frac{1}{(s-2)^2} + \frac{s-1}{s^2-2s-8}$ |
| 7. $f(t) = t \sin at$ | Ans: $\frac{2as}{(s^2+a^2)^2}$ |
| 8. $f(t) = t \cos at$ | Ans: $\frac{a^2-s^2}{(s^2+a^2)^2}$ |
| 9. $f(t) = e^{2t} \cos 3t$ | Ans: $\frac{s-2}{(s-2)^2 + 9}$ |
| 10. $f(t) = t \sin 2t$ | Ans: $-\frac{4s}{(s^2+4)^2}$ |

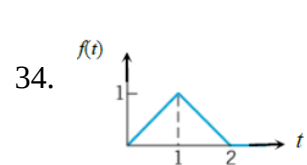
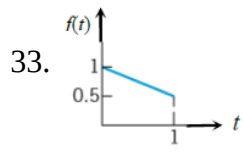
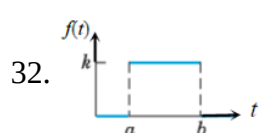
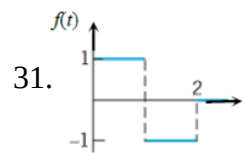
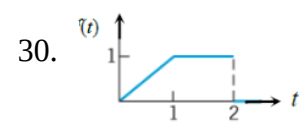
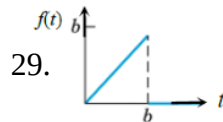
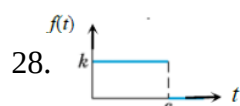
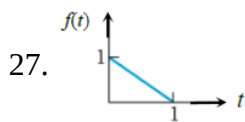
11. $f(t) = t^{10} e^{-7t}$

Ans: $\frac{10!}{(s+7)^{11}}$

12. $f(t) = t \cos bt$

Ans: $\frac{2s^2}{(b^2 + s^2)^2} - \frac{1}{b^2 + s^2}$

III. Defined the function $f(t)$ and find the Laplace transformation of $f(t)$ (27-34) by using definition:



IV. Convert $f(t)$ (35-47) to Unit step function and find the Laplace transformations of $f(t)$:

35. $f(t) = t(0 < t < 2)$

42. $f(t) = t^2 \left(t > \frac{3}{2} \right)$

36. $f(t) = t - 2(t > 2)$

43. $f(t) = \sinh t(0 < t < 2)$

37. $f(t) = \cos 4t(0 < t < \pi)$

44. $f(t) = \sin t \left(\frac{\pi}{2} < t < \pi \right)$

38. $f(t) = e^t \left(0 < t < \frac{\pi}{2} \right)$

45. $f(t) = 16t^2 u \left(t - \frac{1}{4} \right)$

39. $f(t) = \sin \pi t(2 < t < 4)$

46. $f(t) = e^{t/2} u(t - 3)$

40. $f(t) = e^{-\pi t}(2 < t < 4)$

47. $f(t) = u(t - 2\pi) \sin t$

41. $f(t) = t^2(1 < t < 2)$

Additional problem to solve

1. $f(t) = t^2 u(t - 3)$

Ans: $e^{-3s} \left[\frac{2}{s^3} + 6 \frac{1}{s^2} + 9 \frac{1}{s} \right]$

2. $f(t) = e^{-2t} u_{\pi}(t)$

Ans: $e^{-\pi(s+2)} \frac{1}{s+2}$

3. $f(t) = t u_2(t)$

Ans: $e^{-2s} \left(\frac{2}{s} + \frac{1}{s^2} \right)$

4. $f(t) = t u(t - 1)$

Ans: $\frac{e^{-s}}{s} + \frac{e^{-s}}{s^2}$

5. $f(t) = (t - 1) u(t - 1)$

Ans: $\frac{e^{-s}}{s^2}$

6. $f(t) = (t - 1)^2 u(t - 1)$

Ans: $\frac{2 e^{-s}}{s^3}$

7. $f(t) = e^{-2t} u(t - 3)$

Ans: $\frac{e^{-6} e^{-3s}}{s+2}$

8. $f(t) = 4 \cos t u(t - \pi)$

Ans: $-\frac{4s e^{-\pi s}}{s^2 + 1}$

V. Obtain Laplace transformation (48,49) by 1st and 2nd order differentiation:

48. $t \cos 4t$

49. $t^2 e^{-3t}$

Exercise Set 1.2

I. Find the inverse Laplace transformation of the following functions (1-22) assuming a, b, L, n as constants:

- | | | |
|--|---|-----------------------------------|
| 1. $\frac{0.2s+1.8}{s^2+3.24}$ | 12. $\frac{3s+4}{s^2+4s+5}$ | 23. $\frac{3}{s^2+s/4}$ |
| 2. $\frac{5s+1}{s^2-25}$ | 13. $\frac{3s}{s^2-2s+2}$ | 24. $\frac{20}{s^3-2\pi s^2}$ |
| 3. $\frac{s}{L^2 s^2 + n^2 \pi^2}$ | 14. $\frac{\omega \cos \theta + s \sin \theta}{s^2 + \omega^2}$ | 25. $\frac{1}{s(s^2 + \omega^2)}$ |
| 4. $\frac{1}{(s+\sqrt{2})(s-\sqrt{3})}$ | 15. $\frac{6(s+1)}{s^4}$ | 26. $\frac{1}{s^4-s^2}$ |
| 5. $\frac{12}{s^4} - \frac{228}{s^6}$ | 16. $\frac{\pi}{(s+\pi)^2}$ | 27. $\frac{s+1}{s^4+9s^2}$ |
| 6. $\frac{4s+32}{s^2-16}$ | 17. $\frac{6}{(s+1)^3}$ | 28. $\frac{3s+4}{s^4+k^2 s^2}$ |
| 7. $\frac{s+10}{s^2-s-2}$ | 18. $\frac{21}{(s+\sqrt{2})^4}$ | 29. $\frac{1}{s^3+as^2}$ |
| 8. $\frac{1}{(s+a)(s+b)}$ | 19. $\frac{4}{s^2-2s-3}$ | 30. $\frac{2s+1}{(s^2+1)(s-2)}$ |
| 9. $\frac{7.5}{s^2-2s-8}$ | 20. $\frac{\pi}{s^2+10\pi s+24\pi^2}$ | 31. $\frac{s+1}{s^2(s-3)}$ |
| 10. $\frac{\frac{1}{16}}{s^2+s+\frac{1}{2}}$ | 21. $\frac{2s-1}{s^2-6s+18}$ | |
| 11. $\frac{s^2-6.25}{s^2-6.25}$ | 22. $\frac{a(s+k)+b\pi}{(s+k)^2+\pi^2}$ | |

Additional problem to solve

- | | |
|---|---|
| 1. $\frac{s^2+1}{s^3}$ | Ans: $1 + \frac{t^2}{2}$ |
| 2. $\frac{1}{2s-5}$ | Ans: $\frac{1}{2} e^{\frac{5}{2}t}$ |
| 3. $\frac{1}{s^2-16}$ | Ans: $\frac{1}{4} \sinh 4t$ |
| 4. $\frac{2s}{s^2-9}$ | Ans: $2 \cosh 3t$ |
| 5. $\frac{4}{s-2} - \frac{s}{s^2-16} + \frac{4}{s^2-4}$ | Ans: $4e^{2t} - \cosh 4t + 2 \sinh 2t$ |
| 6. $\frac{5}{s} - \frac{3s}{s^2+16} + \frac{2}{s^2+4}$ | Ans: $5 - 3 \cos 4t + \sin 2t$ |
| 7. $\frac{10}{(s+3)^4}$ | Ans: $\frac{10}{6} e^{-3t} t^3$ |
| 8. $\frac{1}{(s-2)^2+1}$, | Ans: $e^{2t} \sin t$ |
| 9. $\frac{s-1}{(s-1)^2+4}$, | Ans: $e^t \cos 2t$ |

10. $\frac{s+2}{(s+2)^2-25}$, **Ans:** $e^{-2t} \cosh 5t$
11. $\frac{1}{(s+3)^2-4}$, **Ans:** $\frac{1}{2}e^{-3t} \sinh 2t$
12. $\frac{s}{s^2+4s+13}$, **Ans:** $e^{-2t} \cos 3t - \frac{2}{3}e^{-2t} \sin 3t$
13. $\frac{2s+1}{s^2+4s+13}$, **Ans:** $2e^{-2t} \cos 3t - e^{-2t} \sin 3t$
14. $\frac{1}{s^2-5s+6}$, **Ans:** $e^{3t} - e^{2t}$
15. $\frac{3s+1}{(s+1)(s^2+1)}$, **Ans:** $-e^{-t} + \cos t + 2 \sin t$
16. $\frac{4s+5}{(s-1)^2(s+2)}$, **Ans:** $\frac{1}{3}e^t + 3te^t - \frac{1}{3}e^{-2t}$
17. $\frac{3s^2+13s+26}{s(s^2+4s+13)}$, **Ans:** $2 + e^{-2t} \cos 3t + e^{-2t} \sin 3t$
18. $\frac{s^3-5s^2+6}{s^2-5s+6}$, **Ans:** $t^3 - 5t + 1$
19. $\frac{s^4+2+4s}{s^2+25}$, **Ans:** $4 \cos 5t + \frac{2}{5} \sin 5t$
20. $\frac{3}{(s+2)^2+9}$, **Ans:** $e^{-2t} \sin 3t$
21. $\frac{5s-7}{s^2-6s+25}$, **Ans:** $5e^{3t} \left(\cos 4t + \frac{2}{5} \sin 4t \right)$
22. $\frac{s}{s^2-6s+10}$, **Ans:** $e^{3t} (\cos t + 3 \sin t)$
23. $\frac{s^2-6s+10}{s+1}$, **Ans:** $\frac{3e^{2t}}{10} - \frac{2e^{-3t}}{15} - \frac{1}{6}$
24. $\frac{6}{s(s-2)(s+3)}$, **Ans:** $e^{4t} - e^{-2t}$
25. $\frac{(s+2)(s-4)}{6s-17}$, **Ans:** $5e^{2t} + e^{3t}$
26. $\frac{s^2-5s+6}{s}$, **Ans:** $e^{-t} - te^{-t}$
27. $\frac{(s+1)^2}{7s^2+14s-9}$, **Ans:** $-40e^t - 12te^t + 47e^{2t}$

II. Find the inverse Laplace transformation (32-37) associated with the unit step function:

32. $e^{-3s}/(s-1)^3$	34. e^{-3s}/s^4	36. $\frac{s+1}{s^2}e^{-s}$
33. $6(1-e^{-\pi s})/(s^2+9)$	35. $2(e^{-s}-e^{-3s})/(s^2-4)$	37. $\frac{2s-10}{s^3}e^{-5s}$
Additional problem to solve		
01. $\frac{e^{-3s}}{s}$ Ans: $u(t-3)$	02. $\frac{e^{-s}}{s^2}$ Ans: $(t-1)u_1(t)$	
03. $\frac{e^{-\pi s}}{s^2+1}$ Ans: $\sin(t-\pi)u_\pi(t)$		

Exercise Set 1.3

Solve the IVPs by the Laplace transformation:

1. $y'' - y = t; y(0) = 1, y'(0) = 1$
2. $y'' + y' + 9y = 0; y(0) = 0.16, y'(0) = 0$
3. $y' + 5.2y = 19.4\sin(2t); y(0) = 0$
4. $y' + 2y = 0; y(0) = 1.5$
5. $y'' - y' - 6y = 0; y(0) = 11, y'(0) = 28$
6. $y'' + 9y = 10e^{-t}; y(0) = 0, y'(0) = 0$
7. $y'' - \frac{1}{4}y = 0; y(0) = 12, y'(0) = 0$
8. $y'' - 6y' + 5y = 29\cos(2t); y(0) = 3.2, y'(0) = 6.2$
9. $y'' + 7y' + 12y = 21e^{3t}; y(0) = 3.5, y'(0) = -10$
10. $y'' - 4y' + 4y = 0; y(0) = 8.1, y'(0) = 3.9$
11. $y'' - 4y' + 3y = 6t - 8; y(0) = 0, y'(0) = 0$
12. $y'' + 0.04y = 0.02t^2; y(0) = -25, y'(0) = 0$
13. $y'' + 3y' + 2.25y = 9t^3 + 64; y(0) = 1, y'(0) = 31.5$
14. $y'' + 4y' + 5y = 50t; y(0) = 5, y'(0) = -5$
15. $y'' + 16y = 4\delta(t - \pi); y(0) = -1, y'(0) = 0$
16. $y'' - y' - 2y = 12u(t - \pi)\sin(\frac{t}{2}); y(0) = 1, y'(0) = -1$
17. $y'' + 4y = \delta(t - \pi) - \delta(t - 2\pi); y(0) = 1, y'(0) = 0$
18. $y'' + 3y' + 2y = 2u(t - 2); y(0) = 0, y'(0) = 0$

Additional problem to solve

19. $\ddot{y}(t) + y(t) = e^t$, $y(0) = 1, \dot{y}(0) = -2$, **Ans:** $y(t) = \frac{e^t}{2} + \frac{1}{2} \cos t - \frac{5}{2} \sin t$.

20. $\dot{y}(t) = 3$; $y(0) = 2$. **Ans:** $y(t) = 3t + 2$.

21. $\dot{y}(t) = 4t$; $y(0) = 1$. **Ans:** $y(t) = 2t^2 + 1$.

22. $\dot{y}(t) = 2t - 1$; $y(0) = 3$. **Ans:** $y(t) = t(t - 1) + 3$.

23. $\dot{y}(t) = t^2$; $y(0) = 4$. **Ans:** $y(t) = \frac{t^3}{3} + 4$.

24. $\dot{y}(t) = e^{2t}$; $y(0) = 2$. **Ans:** $y(t) = \frac{e^{2t}}{2} + \frac{3}{2}$.

25. $\dot{y}(t) + y(t) = 2$; $y(0) = 0$. **Ans:** $y(t) = 2 - 2e^{-t}$.

26. $\ddot{y}(t) = 5$; $y(0) = 1, \dot{y}(0) = 2$. **Ans:** $y(t) = \frac{t(5t+4)}{2} + 1$.

27. $\ddot{y}(t) - 2\dot{y}(t) = \cos t$; $y(0) = 0, \dot{y}(0) = 1$.

Ans: $f(t) = \frac{7}{10} e^{2t} - \frac{2}{5} \left(\sin t + \frac{1}{2} \cos t \right) - \frac{1}{2}$.

28. $\ddot{y}(t) + 3\dot{y}(t) - y(t) = e^t$; $y(0) = \dot{y}(0) = 0$.

Ans: $f(t) = \frac{1}{3} e^t - \frac{1}{3} e^{\frac{-3t}{2}} \left[\cosh\left(\frac{\sqrt{13}}{2} t\right) + \frac{5}{\sqrt{13}} \sinh\left(\frac{\sqrt{13}}{2} t\right) \right]$.

29. $\ddot{y}(t) - 7\dot{y}(t) + 12y(t) = 0$, $y(0) = 2, \dot{y}(0) = 1$.

Ans: $y(t) = -5e^{4t} + 7e^{3t}$.

30. $\ddot{y}(t) + y(t) = \begin{cases} t, & 0 < t < 1 \\ 0, & t > 1 \end{cases}$, $y(0) = 0, \dot{y}(0) = 0$.

Ans: $y(t) = t - \sin t$ if $0 < t < 1$ and $\cos(t - 1) + \sin(t - 1) - \sin t$ if $t > 1$.

Using the Laplace transformation, solve the system of ODEs:

19. $y_1' = -y_1 + 4y_2$, $y_2' = 3y_1 - 2y_2$; $y_1(0) = 3, y_2(0) = 4$

20. $y_1' = 5y_1 + y_2$, $y_2' = y_1 + 5y_2$; $y_1(0) = 1, y_2(0) = -3$

22. $y_1' = 4y_1 + y_2$, $y_2' = -y_1 + 2y_2$; $y_1(0) = 3, y_2(0) = 1$

23. $y_1'' = y_1 + 3y_2$, $y_2'' = 4y_1 - 4e^t$; $y_1(0) = 2, y_1'(0) = 3, y_2(0) = 1, y_2'(0) = 2$

24. $y_1'' = -2y_1 + 2y_2$, $y_2'' = 2y_1 - 5y_2$; $y_1(0) = 1, y_1'(0) = 0, y_2(0) = 3, y_2'(0) = 0$

25. $y_1' = y_2$, $y_2' = -4y_1 + \delta(t - \pi)$; $y_1(0) = 0, y_2(0) = 0$

26. $y_1' = 2y_1 - 4y_2$, $y_2' = y_1 - 3y_2$; $y_1(0) = 3, y_2(0) = 0$

27. $y_1' = 2y_1 + 4y_2$, $y_2' = y_1 + 2y_2$; $y_1(0) = -4, y_2(0) = -4$

Additional problem to solve

$$28. \begin{cases} \frac{dx(t)}{dt} = 2x(t) - 3y(t) \\ \frac{dy(t)}{dt} = y(t) - 2x(t) \end{cases} \text{ subject to } x(0) = 8, y(0) = 3.$$

$$\text{Ans: } \begin{aligned} x(t) &= 5e^{-t} + 3e^{4t} \\ y(t) &= 5e^{-t} - 2e^{4t} \end{aligned}$$

$$29. \begin{aligned} \dot{x} &= y \\ \dot{y} &= 16x; \quad x(0) = 0, y(0) = 4. \end{aligned}$$

$$\text{Ans: } x(t) = \sinh 4t, \quad y(t) = 4 \cosh 4t$$

$$30. \begin{aligned} \dot{x} &= -4y \\ \dot{y} &= x; \quad x(0) = 2, y(0) = 0. \end{aligned}$$

$$\text{Ans: } x(t) = 2 \cos 2t, \quad y(t) = \sin 2t$$

$$31. \begin{aligned} \dot{x} &= 2x + y \\ \dot{y} &= 4x + 2y; \quad x(0) = 1, y(0) = 6. \end{aligned}$$

$$\text{Ans: } x(t) = e^{2t} (\cosh 2t + 3 \sinh 2t), \quad y(t) = e^{2t} (6 \cosh 2t + 2 \sinh 2t).$$

$$32. \begin{aligned} \dot{x} &= 3x + y \\ \dot{y} &= 4x + 3y; \quad x(0) = 3, y(0) = 2. \end{aligned}$$

$$\text{Ans: } x(t) = e^{3t} (3 \cosh 2t + \sinh 2t), \quad y(t) = e^{3t} (2 \cosh 2t + 6 \sinh 2t)$$

System of ODEs using Laplace Transformation:

28. Tank T_1 initially contains 100gal of pure water. Tank T_2 initially contains 100gal of water in which 150lb of salt are dissolved. The inflow into T_1 is 2gal/min from T_2 and 6gal/min containing 6lb of salt from the outside. The inflow into T_2 is 8gal/min from T_1 . The outflow from T_2 is $2 + 6 = 8$ gal/min, as shown in the figure. The mixtures are kept uniform by stirring. Find the salt contents $y_1(t)$ and $y_2(t)$ in T_1 and T_2 , respectively.

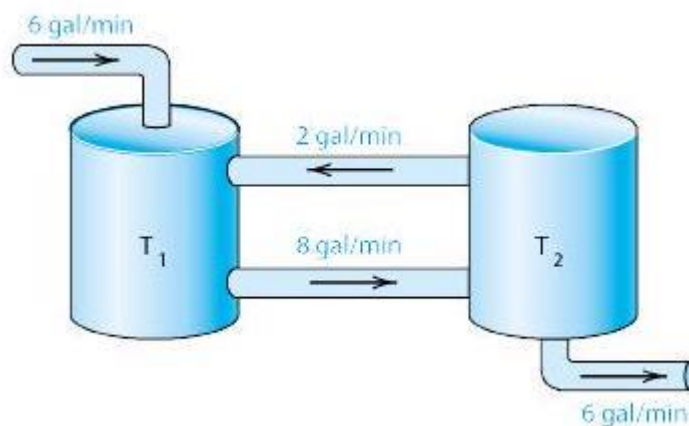


Figure 1: Mixing Problem

29. The mechanical system in Fig. 2 consists of two bodies of mass 1 on three springs of the same spring constant k and of negligibly small masses of the springs. Also damping is assumed to be practically zero. Then the model of the physical system is

$$y_1'' = -ky_1 + k(y_2 - y_1), \quad y_2'' = -k(y_2 - y_1) - ky_2.$$

Here y_1 and y_2 are the displacements of the bodies from their positions of static equilibrium. We regard downward forces as positive and upward as negative. On the upper body, $-ky_1$ is the force of the upper spring and $k(y_2 - y_1)$ that of the middle spring, $y_2 - y_1$ being the net change in spring length. On the lower body, $-k(y_2 - y_1)$ is the force of the middle spring and $-ky_2$ that of the lower spring. Initial conditions: $y_1(0) = 1, y_2(0) = 1, y_1'(0) = \sqrt{3k}; y_2'(0) = -\sqrt{3k}$

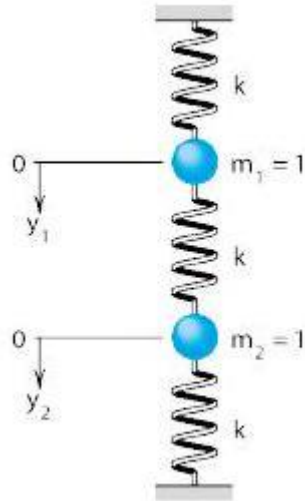


Figure 2: Mass Spring Model

30. Using the Laplace transform and showing the details, find the current $i(t)$ in the circuit in Fig. 3, assuming zero initial current and charge and

$$R = 2\Omega, L = 1H, C = 0.5F, v(t) = 1kV \text{ if } 0 < t < 2 \text{ and } 0 \text{ if } t > 2.$$

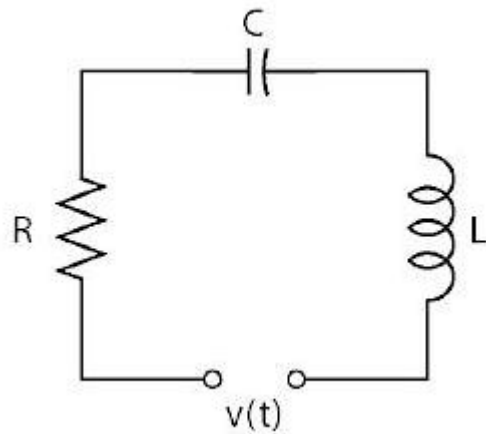


Figure 3: RLC Circuit

Additional problem to solve

31. A resistance R in series with inductance L is connected with e.m.f $E(t) = t$. The current $i(t)$ is given by $L \frac{di}{dt} + Ri = t$; $i(0) = 0$. Use Laplace transform to find the current $i(t)$.

Ans: $i(t) = -\frac{L}{R^2} + \frac{t}{R} + \frac{L}{R^2} e^{-\frac{R}{L}t}$.

32. An inductor of 2 henrys, a resistor of 16 ohms and a capacitor of .02 farads are connected in series with an e.m.f. of E volts. At $t = 0$ the charge on the capacitor and current in the circuit are zero. Find the charge and current at any time $t > 0$ if $E = 300$ (volts).

Ans: $q(t) = 6 - 6e^{-4t} \cos t - 8e^{-4t} \sin 3t$

$i(t) = \frac{dq}{dt} = 50e^{-4t} \sin 3t$.

33. The current $i(t)$ in an electrical circuit is given by the DE, $\frac{d^2i}{dt^2} + 2\frac{di}{dt} = \begin{cases} 0, & 0 < t < 10 \\ 1, & 10 < t < 20 \\ 0, & t > 20 \end{cases}$

$i(0) = \frac{di}{dt}(0) = 0$. Determine current $i(t)$.

Ans: $i(t) = \left[\frac{(t-10)}{2} + \frac{e^{-2(t-10)}}{4} - \frac{1}{4} \right] u(t-10) - \left[\frac{(t-20)}{2} + \frac{e^{-2(t-20)}}{4} - \frac{1}{4} \right] u(t-20)$.

$$= \begin{cases} 0, & 0 < t < 10 \\ \frac{(t-10)}{2} + \frac{e^{-2(t-10)}}{4} - \frac{1}{4}, & 10 < t < 20 \\ 5 + \frac{e^{-2(t-10)}}{4} - \frac{e^{-2(t-20)}}{4}, & t > 20 \end{cases}$$

Exercise Set 2

1. If $z_1 = 2 + 4i$, $z_2 = -3 + 8i$, $z_3 = 2 - 3i$ and $z_4 = -4 - 6i$, then find

- | | |
|---|--|
| a) Locate z_1, z_2, z_3 and z_4 in Argand Diagram | d) $z_1 z_2$ |
| b) $z_1 + z_2$ | e) $\frac{z_3}{z_4} + \frac{z_1}{z_2}$ |
| c) $z_3 - z_1$ | f) $\frac{1}{(z_2)^3} + z_1$ |
| g) $ z_1 \bar{z}_2 - z_3 $ | |

Additional problem to solve

1. If $z_1 = 1 - i$, $z_2 = -2 - 3i$ and $z_3 = 2i$, then evaluate $|3z_1^3 + 2\bar{z}_2 - 5z_3|$.

Ans: $10\sqrt{2}$

2. Write the given number in the form $a + ib$.

- | | | |
|--|---|-----------------------------|
| a. $(5 - 9i) + (2 - 4i)$ | f. $3i + \frac{1}{2 - i}$ | k. $i(5 + 7i)$ |
| b. $i(4 - i) + 4i(1 + 2i)$ | g. $(2 - 3i)(4 + i)$ | l. $\frac{i}{1 + i}$ |
| c. $3(4 - i) - 3(5 + 2i)$ | h. $\left(\frac{1}{2} - \frac{1}{4}i\right)\left(\frac{2}{3} + \frac{5}{3}i\right)$ | m. $\frac{2 - 4i}{3 + 5i}$ |
| d. $\frac{(1 + i)(1 - 2i)}{(2 + i)(4 - 3i)}$ | i. $\frac{(3 - i)(2 + 3i)}{1 + i}$ | n. $\frac{10 - 5i}{6 + 2i}$ |
| e. $\frac{(5 - 4i) - (3 + 7i)}{(4 + 2i) + (2 - 3i)}$ | j. $\frac{(4 + 5i) + 2i^3}{(2 + i)^2}$ | o. $(2 + 3i)^2$ |

Additional problem to solve

- | | |
|------------------------------|----------------------------|
| p) $\frac{(1 + i)^2}{1 - i}$ | q) $\frac{-3 + i}{7 - 3i}$ |
|------------------------------|----------------------------|

3. Find $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$ of $z = \left(\frac{i}{3-i}\right)\left(\frac{1}{2+3i}\right)$.

4. Let $z = -6 - 5i$. Find

a) $\operatorname{Re}\left(\frac{1}{z}\right)$

e) $\operatorname{Re}(iz)$

b) $\operatorname{Re}(z^2)$

f) $|z|$

c) $\operatorname{Im}(2z + 4\bar{z} - 4i)$

g) $(1+i)z$

d) $\operatorname{Im}(\bar{z}^2 + z^2)$

h) $\operatorname{Im}(iz)$

Additional problem to solve

i) Find $\operatorname{Im}\{z\}$, where $z = \frac{(1-i)^2}{1+i}$

j) $\operatorname{Re}\left\{\frac{1+\sqrt{3}i}{1-i}\right\}$

k) $\left|\frac{z}{\bar{z}}\right|$

l) $\operatorname{Im}\left\{\frac{z}{\bar{z}}\right\}$

5. Find the modulus of the following Complex number:

a) $z = (1-i)^2$

c) $z = \frac{2i}{3-4i}$

b) $z = i(2-i) - 4\left(1 + \frac{1}{4}i\right)$

d) $z = \frac{1-2i}{1+i} + \frac{2-i}{1-i}$

6. Find the principal argument of the following Complex numbers:

a) $z = 1+i$

c) $z = \frac{1-i}{(1+i)^2}$

b) $z = -1 + \sqrt{3}i$

d) $z = -5 + 5i$

Additional problem to solve

e) $z = (-1 - i)^4$

g) $z = \frac{(1+i)^3}{(1-i)},$

f) $z = (-2 + 2\sqrt{3}i)^3$

h) $z = \frac{(-1-i)^{12}}{(3+\sqrt{3}i)^6 (-2+2\sqrt{3}i)^3}$

Convert the following Complex numbers into polar form:

a) $z = 2$

e) $z = -\sqrt{3} + i$

b) $z = -1$

f) $z = 1 + i$

c) $z = -3i$

g) $z = -2 - 2\sqrt{3}i$

d) $z = 1 + i$

h) $z = \frac{1+i}{(1-i)^2}$

Additional problem to solve

i) $z = -1 + i$

k) $z = -3 - \sqrt{3}i$

j) $z = \frac{(1-i)^2}{1+i}$

l) $z = \left(\frac{1-i}{1+i}\right)^{18}$

7. Establish the given inequality:

a) If $|z| = 2$, then $|z + 6 + 8i| \leq 13$,

b) If $|z| = 1$, then $1 \leq |z^2 - 3| \leq 4$,

c) Find an upper bound for the modulus of $3z^2 + 2z + 1$ if $|z| \leq 1$,

d) Verify the triangle inequality $|z_1 + z_2| \leq |z_1| + |z_2|$ for $z_1 = 3 - 2i$ and $z_2 = 4 + 2i$.

8. For the following equations, find all values of z and also locate these values in the complex plane:

a) $z = (-1)^{\frac{1}{4}}$

b) $z = (1 + i)^{\frac{1}{5}}$

c) $z = (8)^{\frac{1}{3}}$

d) $z = (-1 + i)^{\frac{1}{5}}$

e) $z = (-1 + \sqrt{3}i)^{\frac{1}{2}}$

f) $z = (-1 - \sqrt{3}i)^{\frac{1}{4}}$

Additional problem to solve

g) $z^2 + 9 = 0$

h) $z^5 - 1 = 0$

i) $z^4 = -81$

j) $z^3 - \sqrt{3} - i = 0$

k) $z^3 = -i$

l) $z^3 - i = 0$

m) $z^3 - 1 + i = 0$

n) $z^3 - 64 = 0$

o) $z^3 + 2 - i2\sqrt{3} = 0$

p) $z^4 - 1 = 0$

9. Describe and graph the locus represented by each of the following:

a) $|z + 3i| > 4$,

b) $1 < |z - 5 + i| \leq 2$,

c) $|z - 1 + 2i| > 5$,

d) $|z| \leq 4$.

Additional problem to solve

(e) $|z + 2i| > 4$.

(f) $1 < |z - 2 + i| \leq 3$.

(g) $\text{Im}\{z^2\} = 9$.

(h) $|z - 1| \leq 1$.

(i) $\text{Re}\{z^2\} < 4$.

(j) $\frac{\pi}{6} \leq \arg z \leq \frac{\pi}{3}$.

(k) $1 < |z + i| \leq 2$

Exercise Set 3

1. Verify Cauchy-Riemann Equations. If possible, find $f'(z)$:

1. $f(z) = z^3$

12. $f(z) = e^{-x} \cos y - i e^{-x} \sin y$

2. $f(z) = 3z^2 + 5z - 6i$

13. $f(z) = x + \sin x \cosh y + i(y + \cos x \sinh y)$

3. $f(z) = \operatorname{Re}(z)$

14. $f(z) = e^{x^2-y^2} \cos 2xy + i e^{x^2-y^2} \sin 2xy$

4. $f(z) = y + ix$

15. $f(z) = 4x^2 + 5x - 4y^2 + 9 + i(8xy + 5y - 1)$

5. $f(z) = 4z - 6\bar{z} + 3$

16. $f(z) = \frac{\cos \theta}{r} - i \frac{\sin \theta}{r}$

6. $f(z) = \bar{z}^2$

17. $f(z) = 5r \cos \theta + r^4 \cos 4\theta + i(5r \sin \theta + r^4 \sin 4\theta)$

7. $f(z) = x^2 + y^2$

18. $f(z) = e^z$

8. $f(z) = \frac{x}{x^2+y^2} + i \frac{y}{x^2+y^2}$

19. $f(z) = e^{-z}$

9. $f(z) = z^5$

20. $f(z) = e^{3z}$

10. $f(z) = \frac{1}{z^7}$

21. $f(z) = 2e^z$

11. $f(z) = 2z^3 + 6z$

Additional problem to solve

22. $f(z) = e^{z^2}$

29. $f(z) = 2ze^z$

23. $f(z) = e^{2z}$

30. $f(z) = 3z^3$

24. $f(z) = 2z^2$

31. $f(z) = z^2$

25. $f(z) = z^9$

32. $f(z) = \frac{1}{z^9}$

26. $f(z) = e^{2x}(\cos 2y + i \sin 2y)$

33. $f(z) = z^{-\frac{2}{3}}$

27. $f(z) = \bar{z}$

34. $f(z) = z^{\frac{3}{5}}$

28. $f(z) = 2z^2 + 3e^z$

35. $f(z) = \frac{1}{z^3}$

Mapping

1. Find the image S' of the square S with vertices at $1 + i$, $2 + i$, $2 + 2i$, and $1 + 2i$ under the linear mapping $T(z) = z + 2 - i$
2. Find the image of the real axis $y = 0$ under the linear mapping-

$$R(z) = \left(\frac{1}{2}\sqrt{2} + \frac{1}{2}\sqrt{2}i \right) z$$

3. Find the image of the circle C given by $|z| = 2$ under the linear mapping $M(z) = 3z$
4. Find the image of the rectangle with vertices $-1 + i$, $1 + i$, $1 + 2i$, and $-1 + 2i$ under the linear mapping $f(z) = 4i \cdot z + 2 + 3i$
5. Find the image of the triangle with vertices 0, 1, and i under the given linear mapping $w = f(z)$:

a. $f(z) = z + 3i$

b. $f(z) = z + 2 - i$

c. $f(z) = 3iz$

d. $f(z) = 1 + iz$

e. $f(z) = 2z - i$

f. $f(z) = 6 - 5iz + 1 - 3i$

g. $f(z) = 3iz + 4$

h. $f(z) = 5e^{i\frac{\pi}{5}}z + 7i$

i. $f(z) = -1 + 2z + 1 - \sqrt{3}i$

j. $f(z) = (3 - 2i)z + 12$

6. Find the image of the triangle with vertices 0, 1, and i under the given linear mapping $w = f(z)$

a. $f(z) = z + 2i$

b. $f(z) = 3z$

c. $f(z) = e^{i\frac{\pi}{4}}z$

d. $f(z) = \frac{1}{2}iz$

e. $f(z) = -3z + i$

f. $f(z) = (1 - i)z - 2$

Complex Integrals

In Problems 1-16, evaluate the given integral along the indicated contour.

1. $\int_C (z + 3) dz$, where C is $x = 2t$, $y = 4t - 1$, $1 \leq t \leq 3$
2. $\int_C (2\bar{z} - z) dz$, where C is $x = -t$, $y = t^2 + 2$, $0 \leq t \leq 2$
3. $\int_C z^2 dz$, where C is $z(t) = 3t + 2it$, $-2 \leq t \leq 2$
4. $\int_C (3z^2 - 2z) dz$, where C is $z(t) = t + it^2$, $0 \leq t \leq 1$
5. $\int_C \frac{z+1}{z} dz$, where C is the right half of the circle $|z| = 1$ from $z = -i$ to $z = i$
6. $\int_C |z|^2 dz$, where C is $x = t^2$, $y = 1/t$, $1 \leq t < 2$
7. $\oint_C \operatorname{Re}(z) dz$, where C is the circle $|z| = 1$
8. $\oint_C \left(\frac{1}{(z+i)^3} - \frac{5}{z+i} + 8 \right) dz$, where C is the circle $|z+i| = 1$, $0 \leq t \leq 2\pi$
9. $\int_C (x^2 + iy^3) dz$, where C is the straight line from $z = 1$ to $z = i$
10. $\int_C (x^2 - iy^3) dz$, where C is the lower half of the circle $|z| = 1$ from $z = -1$ to $z = 1$
11. $\int_C e^z dz$, where C is the polygonal path consisting of the line segments from $z = 0$ to $z = 2$ and from $z = 2$ to $z = 1 + \pi i$
12. $\int_C \sin z dz$, where C is the polygonal path consisting of the line segments from $z = 0$ to $z = 1$ and from $z = 1$ to $z = 1 + i$
13. $\int_C \operatorname{Im}(z - i) dz$, where C is the polygonal path consisting of the circular arc along $|z| = 1$ from $z = 1$ to $z = i$ and the line segment from $z = i$ to $z = -1$

14. $\int_C dz$, where C is the left half of the ellipse $\frac{1}{36}x^2 + \frac{1}{4}y^2 = 1$ from $z = 2i$ to $z = -2i$
15. $\oint_C ze^z dz$, where C is the square with vertices $z = 0$, $z = 1$, $z = 1 + i$, and $z = i$
16. $\int_C f(z) dz$, where $f(z) = \begin{cases} 2, & x < 0 \\ 6x, & x > 0 \end{cases}$ and C is the parabola $y = x^2$ from $z = -1 + i$ to $z = 1 + i$

In Problems 17–20, evaluate the given integral along the contour C given in Figure 5.21.

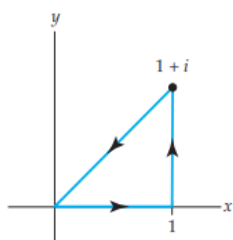


Figure 5.21 Figure for Problems 17–20

17. $\oint_C x dz$

18. $\oint_C (2z - 1) dz$

19. $\oint_C z^2 dz$

20. $\oint_C \bar{z}^2 dz$

In Problems 21–24, evaluate $\int_C (z^2 - z + 2) dz$ from i to 1 along the contour C given in the figures.

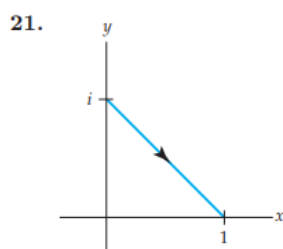


Figure 5.22 Figure for Problem 21

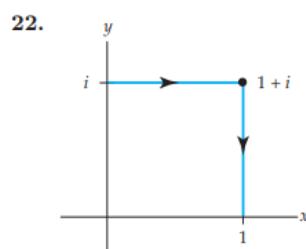


Figure 5.23 Figure for Problem 22

23.

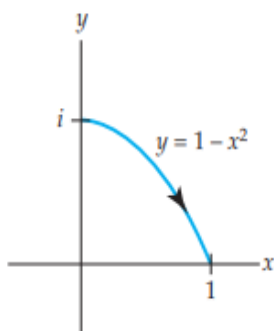


Figure 5.24 Figure for Problem 23

24.

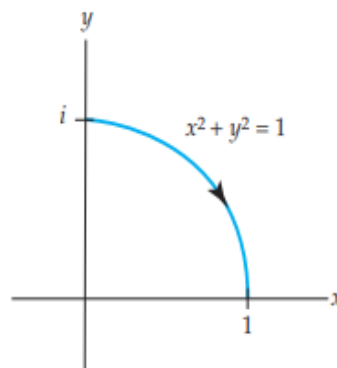


Figure 5.25 Figure for Problem 24